The Fundamental Theorem of Calculus

ANSWER KEY

Section 1: FTC Part 1 (Differentiating Integrals)

1 Find $\frac{d}{dx}[\int_2^x \sqrt{t^3+1}\,dt]$.

a. $\sqrt{x^3+1}$ ✓

b. $\frac{3x^2}{2\sqrt{x^3+1}}$

c. $\sqrt{x^3+1}-3$

d. x^3+1

By FTC Part 1, $\frac{d}{dx}[\int_a^x f(t)dt] = f(x)$. Here it equals $\sqrt{x^3+1}$.

2 Find $\frac{d}{dx}[\int_0^{x^2} \cos(t)dt]$.

a. $2x\cos(x^2)$ ✓

b. $\cos(x^2)$

c. $-2x\sin(x^2)$

d. $\sin(x^2)$

By FTC Part 1 with the Chain Rule: $\cos(u)\cdot\frac{du}{dx} = \cos(x^2)\cdot 2x = 2x\cos(x^2)$.

3 Find $\frac{d}{dx}[\int_{2x}^{3x} t^2 dt]$.

a. $19x^2$ ✓

b. $27x^2 - 8x^2$

c. $35x^2$

d. $5x^2$

$\frac{d}{dx}[\int_{2x}^{3x} t^2 dt] = (3x)^2\cdot 3 - (2x)^2\cdot 2 = 9x^2(3) - 4x^2(2) = 27x^2 - 8x^2 = 19x^2$.

Section 2: FTC Part 2 (Evaluation Theorem)

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4 Evaluate $\int_0^{\pi} \sin x dx$.

a. 2 ✓

b. 0

c. 1

d. -2

Antiderivative is $[-\cos x]_0^{\pi} = -\cos(\pi) - (-\cos 0) = -(-1) - (-1) = 1 + 1 = 2$.

5 Evaluate $\int_1^4 \frac{1}{\sqrt{x}} dx$.

a. 2 ✓

b. 1

c. 4

d. $\frac{1}{2}$

$\int x^{-1/2} dx = [2\sqrt{x}]_1^4 = 2(2) - 2(1) = 4 - 2 = 2$.

6 Evaluate $\int_0^1 e^{2x} dx$.

a. $\frac{e^2 - 1}{2}$ ✓

b. $e^2 - 1$

c. $\frac{e^2}{2}$

d. $2(e^2 - 1)$

$[\frac{1}{2}e^{2x}]_0^1 = \frac{1}{2}(e^2 - e^0) = \frac{e^2 - 1}{2}$.

Section 3: Integration by u-Substitution

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7 Evaluate $\int 2x(x^2+1)^4 dx$.

a. $\frac{(x^2+1)^5}{5}+C$ ✓

b. $\frac{(x^2+1)^5}{10}+C$

c. $(x^2+1)^5+C$

d. $\frac{2(x^2+1)^5}{5}+C$

Let $u=x^2+1 \Rightarrow du=2xdx$. $\int u^4 du=\frac{u^5}{5}+C=\frac{(x^2+1)^5}{5}+C$.

8 Evaluate the definite integral $\int_0^1 xe^{x^2} dx$.

a. $\frac{e-1}{2}$ ✓

b. $e-1$

c. $\frac{e}{2}$

d. $2(e-1)$

Let $u=x^2 \Rightarrow du=2xdx$. When $x=0$, $u=0$; when $x=1$, $u=1$. Integral becomes $\frac{1}{2}\int_0^1 e^u du=\frac{1}{2}[e^1-e^0]=\frac{e-1}{2}$.

9 Evaluate $\int_0^{\pi/4} \tan x \sec^2 x dx$.

a. $\frac{1}{2}$ ✓

b. 1

c. $\frac{1}{4}$

d. $\frac{\sqrt{2}}{2}$

Let $u=\tan x \Rightarrow du=\sec^2 x dx$. Limits: $\tan(0)=0, \tan(\pi/4)=1$. $\int_0^1 u du=[\frac{u^2}{2}]_0^1=\frac{1}{2}$.

Section 4: Average Value of a Function

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10 What is the average value of $f(x) = 3x^2 - 2x$ on the interval $[0, 3]$?

a. 6 ✓

b. 9

c. 18

d. 3

$f_{\text{avg}} = \frac{1}{3-0}\int_0^3 (3x^2 - 2x)dx = \frac{1}{3}[x^3 - x^2]_0^3 = \frac{1}{3}(27 - 9) = \frac{1}{3}(18) = 6.$

11 By the Mean Value Theorem for Integrals, what is guaranteed for a continuous function f on $[a, b]$?

a. **There exists** $c \in (a, b)$ **such that** $f(c) = f_{\text{avg}}$ ✓

b. $f'(c) = 0$

c. $f(a) = f(b)$

d. $f_{\text{avg}} = 0$

The MVT for Integrals guarantees at least one point c where the function value equals its average value: $f(c) = \frac{1}{b-a}\int_a^b f(x)dx$.

12 What is the average value of $f(x) = \sin x$ on $[0, \pi]$?

a. $\frac{2}{\pi}$ ✓

b. $\frac{1}{\pi}$

c. 0

d. 1

$f_{\text{avg}} = \frac{1}{\pi-0}\int_0^\pi \sin x dx = \frac{1}{\pi}[-\cos x]_0^\pi = \frac{1}{\pi}(1 - (-1)) = \frac{2}{\pi}.$

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